

**NEW JERSEY DEPARTMENT OF TRANSPORTATION
ROUTE 49 BRIDGE OVER MAURICE RIVER
BRIDGE SUPERSTRUCTURE REPLACEMENT
CITY OF MILLVILLE, CUMBERLAND COUNTY, NJ
CONTRACT NO. 036153140, FEDERAL PROJECT NO. NHP-0049(303)**

**CONSTRUCTION SCHEDULE – DESIGNER NARRATIVE
PS&E SUBMISSION – June 2024**

Anticipated Production Rates

The anticipated production rates used for the construction schedule were primarily based on the NJDOT Construction Scheduling Manual, 2013. In general, the items are estimated as a Type 2 (Reconstruction or Upgrading Existing Highways) Project. The production rates assumed in the development of the construction schedule are based on Table A and Table C in Section 7.6 of the above referenced manual. These tables are based on an 8-hour working day per crew. The schedule is based on one work crew per activity.

Anticipated Workforce

This project is expected to require the use of nine different crew types to handle the various types of construction operations needed for the project. A listing of the anticipated crews and their trade makeup is as follows:

1. Construction Layout Crew (1) (3 man survey crew)
2. Grading Crews (1) (8 – Operating Engineers / Oiler / Grade Foreman / Truck Drivers)
3. Curb/Sidewalk/Misc. Concrete Crews (2) (3 – Form Setter / Concrete Finishers)
4. Bituminous Paving Crew (1) (6 – Operating Engineers / Oiler / Screedman / Foreman / Rakers)
5. Landscape and Site Mitigation Work (1) (3 – Operating Engineers / Oilers / Foreman)
6. Electrical Work (1) (3 – Operating Engineer / Electricians / Electrical Foreman)
7. Guide Rail (1) (3 – Operating Engineer / Oilers / Ironworkers)
8. ITS Work (1) (3 man crew – Operating Engineer / Electricians / Laborers)
9. Bridge Crew (1) (20 – Operating engineers / foreman / oiler / carpenters / ironworkers / welders / concrete finishers)
10. Drainage/Utility Work (1) (5 – Operating Engineers / foreman / truck driver / pipe layer)

Anticipated Winter Season Work

The construction duration, from the construction start date to substantial completion, is expected to be approximately 22 months, containing two winter seasons. The construction schedule was developed using work specific calendars for roadway/landscape work and bridge work. The number of working days for each calendar month is consistent with Appendix B of the NJDOT Construction Scheduling Manual, 2013 and Table 108.11.01-1 of the 2019 Standard Specifications.

Work associated with construction of asphalt pavement, landscaping, curb, and driveways were not assigned the roadway/landscape calendar to restrict winter work. Based on current schedule, no winter work is anticipated for roadway/landscape. The construction of the bridge deck, relief slabs; concrete repairs of abutments, piers and wingwalls were assigned the bridge calendar to limit winter working days.

Permit Requirements

NJDOT Bureau of Landscape Architecture and Environmental Solutions currently anticipates the following permits:

- U.S. Army Corps of Engineers (USACE) Nationwide Permits 23
- U.S. Coast Guard Bridge Permit
- NJDEP CAFRA/Waterfront Development (In-water and Upland)/Water Quality Certificate

- NJDOT Statewide Permit (0000-05-0031.2 LUP220001) for a NJDEP Coastal Zone Management (CZM) General Permit (GP) 23
- NJDEP TWA (Treatment Works Approval)

A restriction on in-water work from March 1st to June 30th and October 1st to November 30th is expected to protect waters with fishery resources. This will be confirmed during the permit process. Cofferdams will be installed outside the restricted period, such that work can be conducted within these enclosures during the restricted period. Cofferdam will be installed to maintain a minimum of 50% of water passage.

A restriction on tree clearing from April 1st to August 30th might be imposed to avoid injuring or killing nesting migratory birds. This will be confirmed during the permit process.

Utility Requirements

There are four utility companies with facilities within the project limits. Project improvements require temporary structure utilities support and replacement of Verizon underground telecommunication facilities, City of Millville water and sewer facilities relocations, South Jersey Gas gas main relocations, and installation of Atlantic City Electric lighting facilities/underground electric signal conduit. The bridge stage construction and operation of equipment will require relocation of some of these facilities during specific stages and times of the project. Prior to the start of Traffic Control Stage 2, the proposed relocation of the existing sanitary sewer force main must be completed.

During Pre-Stage 1A, two existing water valves within the Rte. 49 EB bridge sidewalk must be reset flush with proposed temporary pavement due to three feet of sidewalk being removed to make room for two lanes during traffic control (these valves to be buried and abandoned once existing 12" water main & irrigation stub are abandoned during Stage 2).

During Stage 1 bridge construction, Verizon telecommunication conduits will be temporarily supported. Supports will be removed once Verizon conduits can be attached to new superstructure. All Verizon telecommunication conduits/cables must be installed prior to installation of conflicting Stage 1 drainage features (refer to contract Drainage-Utility Plans). South Jersey Gas (SJG) gas mains are located off bridge; proposed gas main will be installed by SJG's contractor and must be installed accordingly around proposed drainage features as shown on plans. Proposed water main on bridge and a portion of proposed water main under roadway will be installed (cap end). Existing sanitary manhole will be removed, and existing abandoned sewer line will be capped/filled. Proposed 2-3" R.M.C., Lighting, and Junction boxes on Rt49 WB side of bridge will be installed (refer to contract Traffic Signal Plans).

During Stage 2 bridge construction, after installation drainage features, remaining portion of water main/services under roadway will be installed (connected to portion placed during Stage 1) and tied into existing water main/services. Portions of existing water main and sanitary sewer force main that have been relocated will be removed (refer to contract Drainage-Utility Plans). Proposed 2-3" R.M.C., Lighting, and Junction boxes on Rt49 EB side of bridge will be installed (refer to contract Traffic Signal and Electrical Plans).

During Post Stage 2 bridge construction, the remaining portion of the Rte. 49 WB sidewalk will be constructed. All remaining water valves, gas valves, telecommunications manholes, and sanitary sewer manholes will be reset prior to final mill and overlay with surface course.

Refer to the contract Drainage-Utility Plans, Traffic Control Plans, Traffic Signal Plans, and Construction Details. The following is a list of Utility Owners within the project limits:

Owner	Facility
Verizon of New Jersey, Inc.	Telecommunication
City of Millville Water and Sewer	Water/Sewer
South Jersey Gas	Gas
Atlantic City Electric	Metered Lighting, Underground Electric Signal Conduit

Right-of-Way Requirements

There are several ROW acquisitions required for this project. They are comprised of two Tidelands easements, five temporary work areas and one utility easement. One Tidelands easement, TE88A, is required for bridge construction and the other, TE88B, is needed for the sanitary sewer construction. The five temporary work areas, E86A, E86B, E86C, E86D and E86E, are on Green Acres encumbered City of Millville property. They are needed to upgrade guiderail and access the construction of the bridge and sanitary sewer. The utility easement UE86F is on City of Millville property which is not encumbered by Green Acres. Please refer to the ROW Plans for additional information. It is anticipated that all ROW will be acquired prior to the start of construction.

Community Commitments

The Department is committed to maintaining access for vehicular traffic throughout the project area during construction. No specific time commitments to the community were made regarding completion of construction.

Lead Time for Special Materials

Construction of the project will require lead time for some activities. Some activities involving the production of working drawings have a lead time of 30 to 45 days. Refer to the preliminary construction schedule for further information on lead time. No special materials will be used in this project. Therefore, there is no additional lead time for special materials.

Detour

As depicted on the Traffic Control and Staging Plans, the project will be constructed without implementing any long term detours. During pre-stage 1B, stage 1 and 2, temporary short term lane closures of either the eastbound or westbound lane are expected for the staging sheeting installation and beam erection operations. A temporary short term lane closure is expected for these operations to accommodate crane placement. Overnight shut down or short term detour for the Ware Avenue near the intersection with Rte.49 may be used to perform a drainage re-construction.

Critical Milestones

A summary of critical milestones is as follows:

Construction Start Date	November 21,2024
Pre-Stage 1A Complete	January 21, 2025
Pre-Stage 1B Complete	January 29, 2025
Stage 1 Complete	September 11, 2025
Stage 2 Complete	July 15, 2026
Post-Stage 2 Complete	August 5, 2026
Substantial Completion	August 5, 2026
Completion	October 28, 2026

Anticipated Schedule Problems

Anticipated problems may include the following:

- Advanced relocation of sanitary sewer line is required prior to Stage 2 bridge demolition. Stage 2 bridge work could be delayed if sanitary sewer installation encounters unforeseen conditions during its installation.
- Problems caused by lack of communication, delays, and conflicts with the utility companies.

Project Schedule Acceleration

Based on the proposed schedule of construction, the Contractor can reach substantial completion in less than 22 months. Accelerating the schedule may be possible by working more than one shift or multiple crews. Accelerating the schedule to allow substantial completion in less than one construction season would be extremely challenging due to the staging sequence and environmental restrictions.